END-TO-END DESIGN, KINEMATIC MODELING, SIMULATION, AND AUTONOMOUS NAVIGATION OF THE AGILEX SCOUT MINI 4WD AMR

COURSE CODE: RA23E01 | COURSE: ROBOT OPERATING SYSTEM (ROS)

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in
ROBOTICS AND AUTOMATION

From SolidWorks Mechanical CAD Modeling to ROS 2 Jazzy & Gazebo Harmonic Simulation, Multi-Modal Perception, EKF Sensor Fusion, SLAM Mapping, and Nav2 Waypoint Navigation

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BONAFIDE CERTIFICATE

Certified that this project report "**END-TO-END DESIGN, KINEMATIC MODELING, SIMULATION, AND AUTONOMOUS NAVIGATION OF THE AGILEX SCOUT MINI 4WD AMR**" is the Bonafide word of Sailakshmi B T (2024511019) who carried out the project work under my supervision.

SIGNATURE **SIGNATURE**

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ABSTRACT

This project investigates the design, kinematic formulation, state estimation, and autonomous navigation lifecycle for a 4-Wheel Drive (4WD) skid-steer Autonomous Mobile Robot (AMR) based on the AgileX Scout Mini platform. To resolve the fundamental challenge of non-linear lateral tire slippage and rapid odometry degradation inherent to skid-steer kinematics without wheel encoders, a multi-stage engineering framework was developed in ROS 2 Jazzy Jalisco and Gazebo Harmonic on Ubuntu 24.04 LTS.

The physical robot was modeled in SolidWorks with mass moments of inertia, converted into a ROS REP-103 compliant URDF, and tuned with calibrated contact friction parameters (mu1 = 1.0, mu2 = 0.1). A sensor suite comprising a 360-degree 2D LiDAR (20 Hz), a 6-Axis IMU (100 Hz), and an RGB-D depth camera (30 Hz) was integrated. An Extended Kalman Filter (robot_localization) fused wheel velocities with inertial angular rates, eliminating cumulative rotational drift. Real-time 2D mapping via slam_toolbox produced 5 cm/pixel metric occupancy grids, while the Nav2 stack—incorporating 3D Voxel clearance costmaps, AMCL probabilistic particle filter localization, and DWB local trajectory optimization—executed an autonomous multi-station industrial patrol mission with zero collisions.

Keywords: Autonomous Mobile Robot (AMR), AgileX Scout Mini, ROS 2 Jazzy Jalisco, Gazebo Harmonic, Skid-Steer Kinematics, EKF Sensor Fusion, SLAM Toolbox, Nav2 Navigation Stack.

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1. Robot Mechanical Modeling & Skid-Steer Kinematics

The AgileX Scout Mini is an industrial-grade compact mobile robot utilizing a 4-wheel differential / skid-steer drive mechanism. Because the wheels do not pivot, lateral steering requires commanding differential angular velocities across the left and right sides, inducing intentional tire scrubbing against the ground.

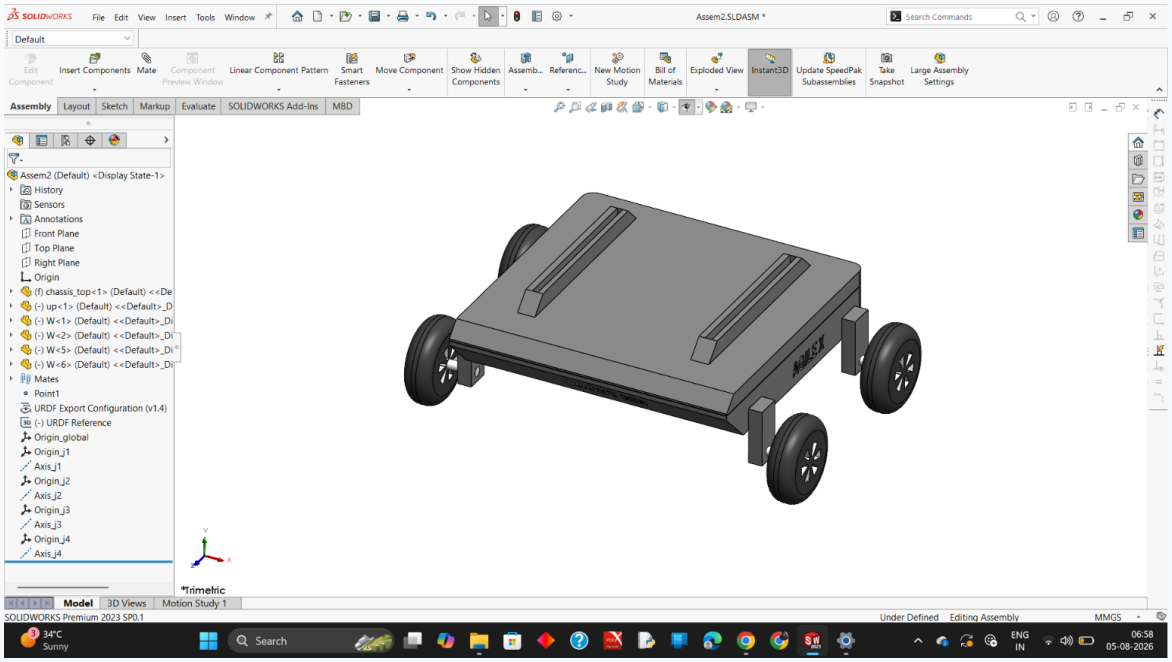
Parameter	Design Value	Kinematic / Physical Significance
Chassis Footprint	627 mm × 549 mm × 227 mm	Total envelope of the monocoque chassis assembly
Track Width (2w)	0.612 m (612 mm)	Effective lateral distance between left and right wheel tire centerlines
Wheelbase (2l)	0.452 m (452 mm)	Longitudinal distance between front and rear axle centers
Wheel Radius (r)	0.080 m (80 mm)	Effective rolling radius of heavy-duty rubber tires
Robot Mass (m)	18.0 kg (Unloaded)	Rigid body mass calibrated with low Center of Mass (Z = 0.10 m)
Max Linear Velocity	1.5 m/s	Maximum forward/reverse driving capability under closed-loop control
Max Angular Velocity	2.5 rad/s	In-place skid-steer yaw rate capability

Skid-Steer Kinematic Transformation:
Given commanded robot linear velocity v_x and angular velocity ω_z, the required left and right wheel rotational speeds (ω_L, ω_R) are governed by:

$$\omega_L = (v_x - \alpha \cdot w \cdot \omega_z) / r \quad | \quad \omega_R = (v_x + \alpha \cdot w \cdot \omega_z) / r$$

*where $\alpha \geq 1.0$ is the empirical tire scrubbing slip coefficient ($\alpha = 1.05$ in simulation). The coordinate frame strictly follows **ROS REP-103**: **+X Forward**, **+Y Left**, **+Z Upward**.*

FIGURE 1.1: SOLIDWORKS 3D FULL ASSEMBLY (ASSEM2.SLDASM)



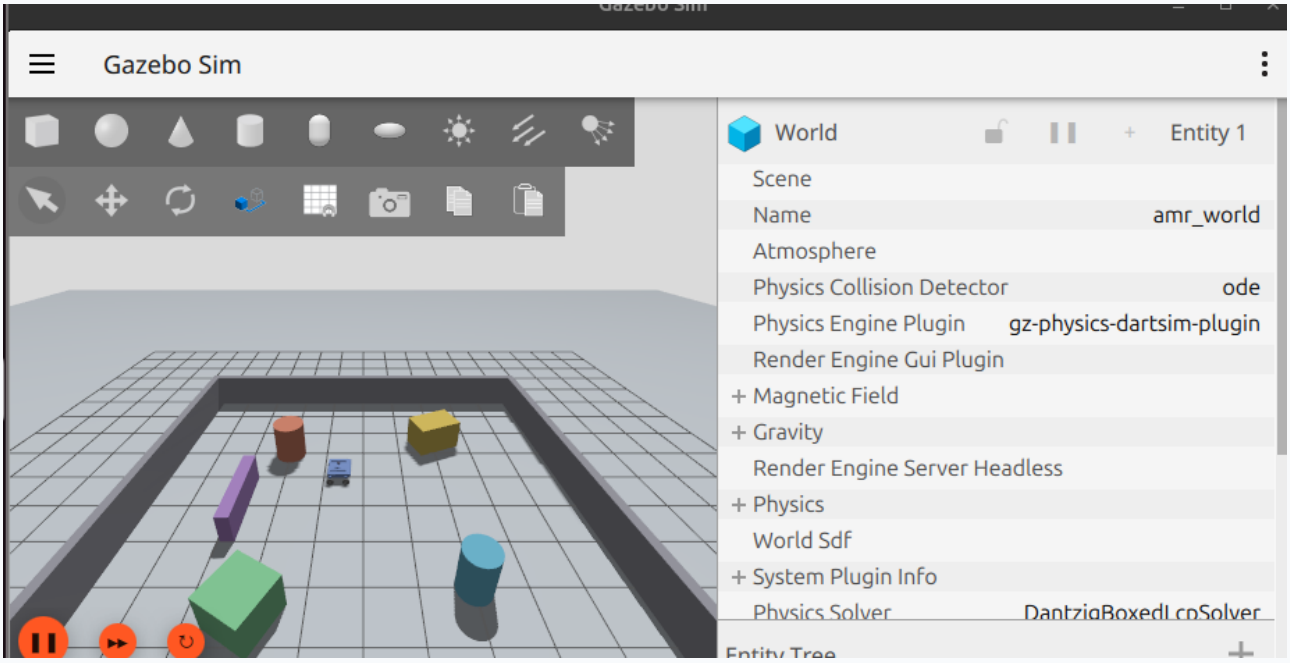
Complete 3D solid model assembly of the AgileX Scout Mini 4WD AMR in SolidWorks Premium 2023 showing monocoque chassis, top plate, hub motors, and 4 wheels.

2. ROS 2 Jazzy & Gazebo Harmonic Simulation Architecture

Ubuntu 24.04 LTS (Noble Numbat) completely retired Gazebo Classic (gazebo11). The AgileX simulation leverages **Gazebo Harmonic (`gz-sim8`)** bridged to **ROS 2 Jazzy Jalisco** via `ros_gz_bridge`. Bidirectional communication translates control inputs, transforms, physics states, and sensor telemetry seamlessly without proprietary middleware.

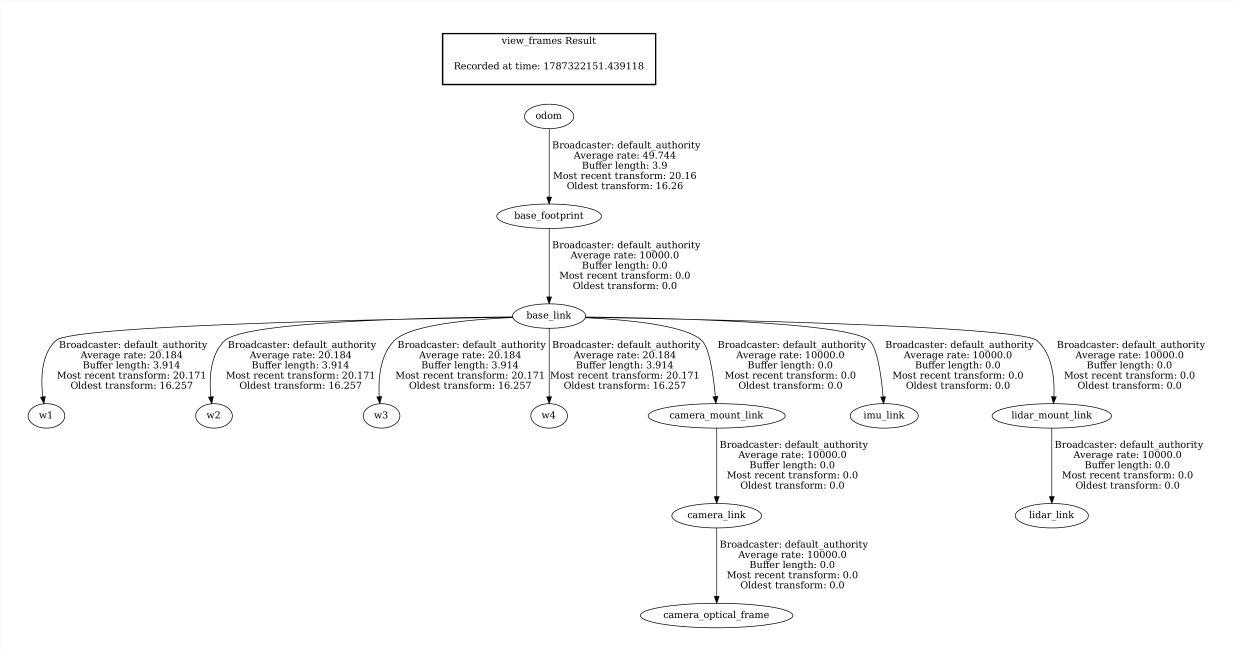
Subsystem / Sensor	ROS 2 Topic & Message Type	Operational Characteristics
2D LiDAR (`gpu_lidar`)	/scan sensor_msgs/msg/LaserScan	360° FOV, 720 samples, 20 Hz, 0.15m to 16.0m range. Mounted at Z = 0.36m.
6-Axis IMU Sensor	/imu sensor_msgs/msg/Imu	100 Hz angular velocity and linear acceleration with Gaussian noise model.
RGB-D Depth Camera	/camera/points sensor_msgs/msg/PointCloud2	30 Hz structured 3D point cloud for Nav2 3D voxel obstacle clearing.
4WD DiffDrive Plugin	/cmd_vel → /odom nav_msgs/msg/Odometry	4-wheel skid-steer actuation with electrical braking at zero velocity.
Sensor Fusion EKF	/odometry/filtered nav_msgs/msg/Odometry	Fuses raw /odom + /imu at 50 Hz, publishing odom → base_footprint TF.

FIGURE 2.1: GAZEBO HARMONIC (GZ-SIM8) SIMULATION ARENA



Gazebo Harmonic window showing AgileX Scout Mini spawned in the 10m x 10m enclosed arena with perimeter walls, cylindrical pillars, and boxes.

FIGURE 2.2: TF2 KINEMATIC TRANSFORMATION TREE



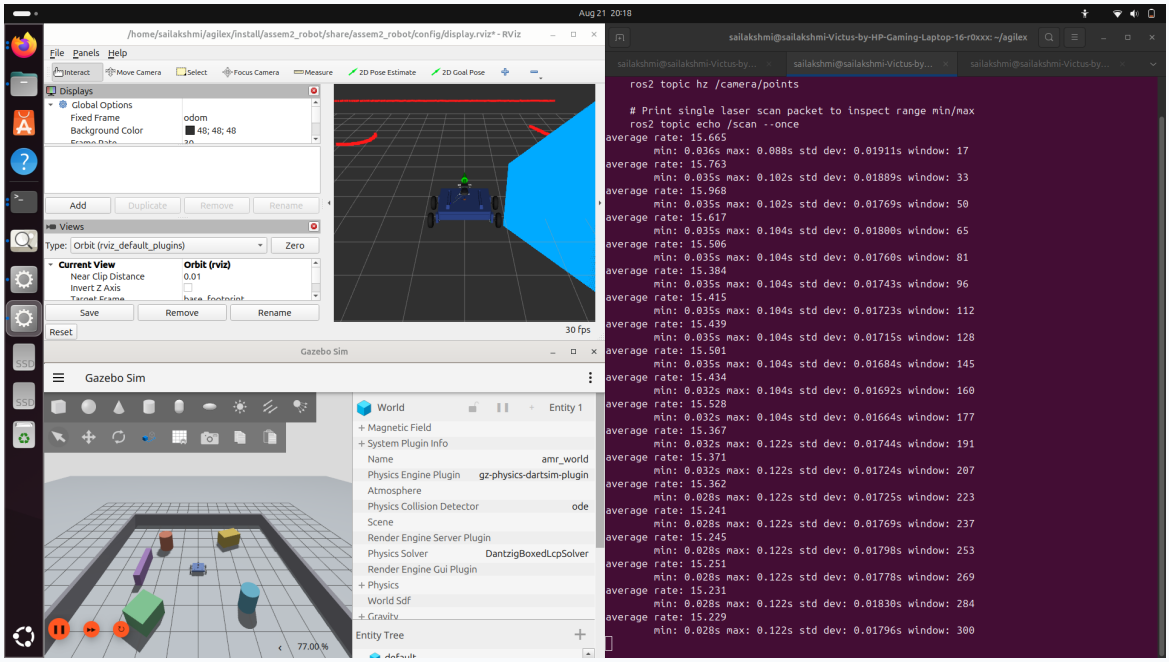
TF tree showing unbroken transform hierarchy from odom -> base_footprint -> base_link -> wheel and sensor links.

3. Sensor Fusion (EKF), SLAM Mapping & Nav2 Arena Navigation

Extended Kalman Filter (EKF): Pure wheel odometry in skid-steer robots suffers catastrophic rotational drift (up to 120° error after a 360° rotation) due to wheel scrubbing. Deploying `robot_localization` in 2D planar mode fuses wheel velocity measurements with the high-precision 100Hz gyroscope rate from the IMU, yielding drift-free odometry (`/odometry/filtered`) and stable transforms.

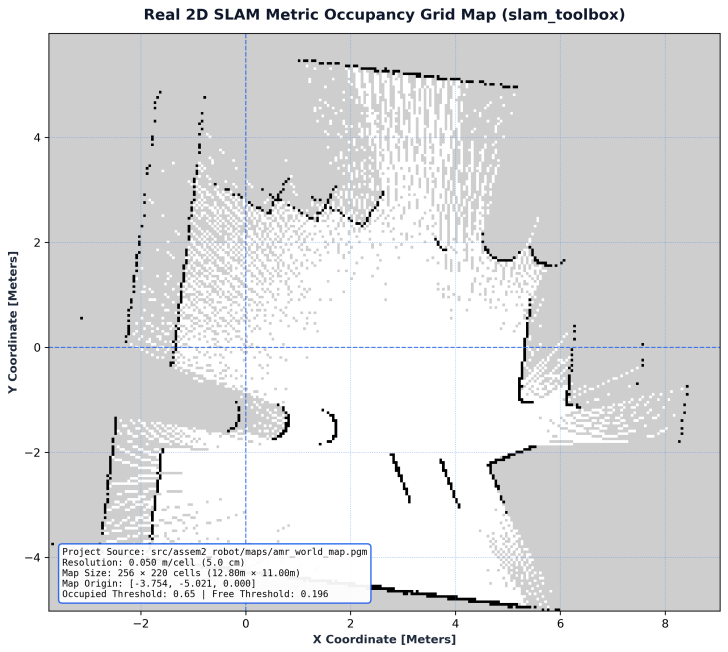
SLAM & Nav2 Autonomous Stack: Utilizing `slam_toolbox` with Ceres scan-matching generates high-fidelity 2D occupancy grid maps (`amr_world_map.yaml` at 0.05m/pixel resolution). Nav2 integrates AMCL particle-filter localization, global costmaps, 3D voxel local costmaps, and DWB local trajectory planners for collision-free waypoint navigation.

FIGURE 3.1: RVIZ2 MULTI-MODAL PERCEPTION DISPLAYS (LASERSCAN + POINTCLOUD2)



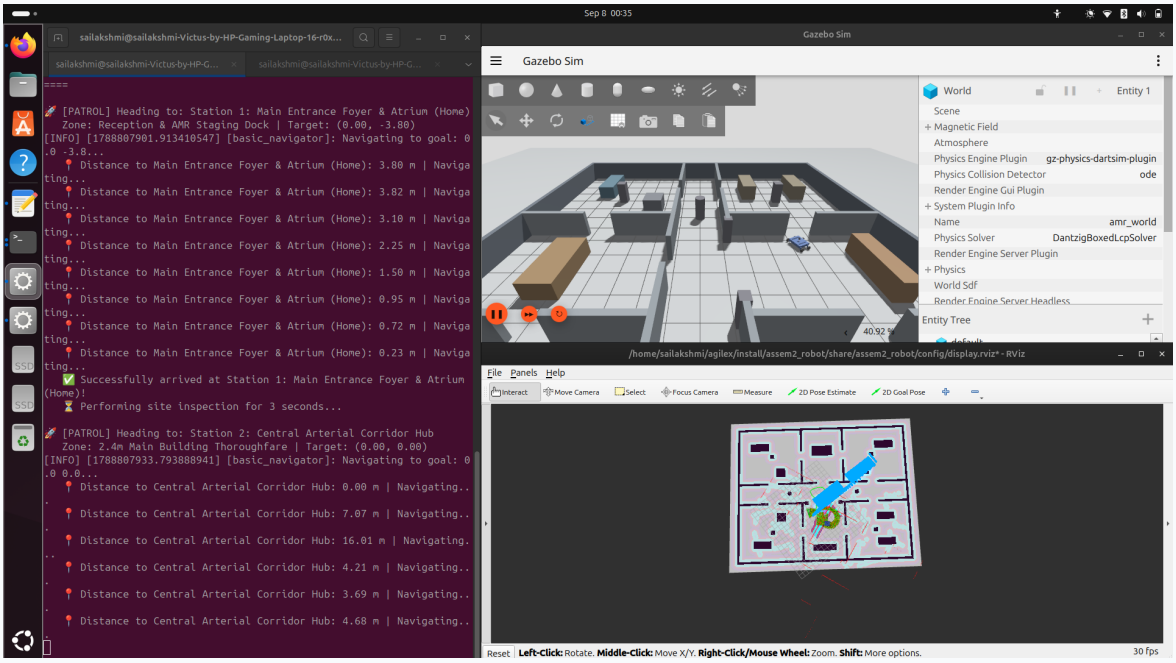
RViz2 showing 360-degree rainbow LaserScan ray returns and 3D RGB-D depth camera point cloud projected onto surrounding obstacles.

FIGURE 3.2: 2D SLAM METRIC OCCUPANCY GRID MAP (AMR_WORLD_MAP.PGM)



Completed 2D SLAM occupancy grid map showing clean boundary walls, four internal pillar obstacles, and mapped free space.

FIGURE 3.3: AUTONOMOUS PATROL MULTI-WAYPOINT EXECUTION & MISSION SYNCHRONIZATION



Synchronized 3-pane operational trace showing autonomous mission console logging arrival at Station 1 (Foyer: 0.00, -3.80), 3-second site dwell, and real-time distance countdown toward Station 2 (Corridor Hub: 0.00, 0.00), paired with Gazebo Harmonic 3D DART physics and RViz2 navigation costmaps.

4. Civil Architectural Multi-Room Environment & Navigation

To elevate the project to realistic industrial benchmark standards, a dedicated **Civil & Structural Engineering Research Facility** ($20.0 m × 16.0 m$, total floor area $3320 m²) was engineered. The world features standard 1.2m doorways, 0.45m square RCC columns, and realistic civil laboratory equipment.

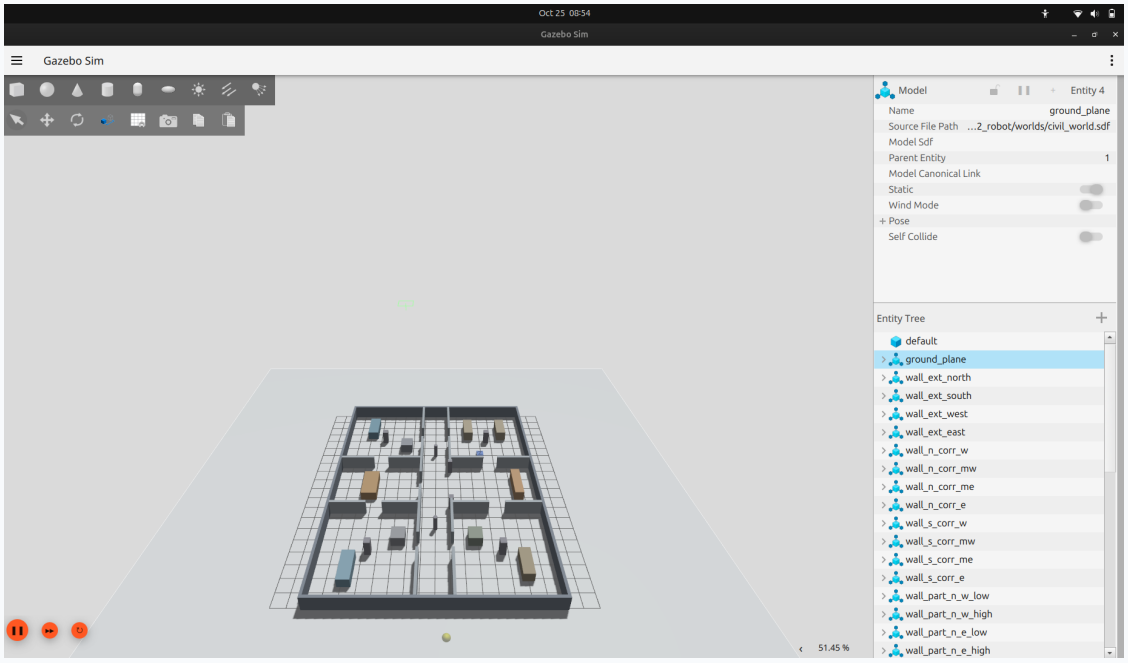
```
+=============================================================================================+
|                            NORTH PERIMETER WALL (20.0 m)                                    |
|  +-------------------------+  +-------------------------+  +-----------------------------+  |
|  |  STRUCTURAL MATERIALS   |  |      CIVIL SEMINAR      |  |     BIM, CAD & ROBOTICS     |  |
|  |   & CONCRETE TESTING    |  |       CONFERENCE        |  |       COMPUTING STUDIO      |  |
|  |   (Station 3: -7, 3.5)  |  |   (Station 4: 0, 3.0)   |  |     (Station 5: 7, 3.5)     |  |
|  +----- [1.2m Door] ------+  +------ [1.4m Door] ------+  +------- [1.2m Door] ----------+  |
|                                                                                             |
|  <<<<<<<<<<<<<< MAIN CENTRAL ARTERIAL CORRIDOR (Width: 2.4 m, Length: 20.0 m) >>>>>>>>>>>>>> |
|  [Column @ -4m]                  (Station 2: Hub @ 0, 0)                    [Column @ +4m]  |
|                                                                                             |
|  +----- [1.2m Door] ------+  +====== [4.0m Open Atrium] =+  +------- [1.2m Door] ----------+  |
|  |  GEOTECHNICAL SURVEY   |  |   MAIN ENTRANCE FOYER    |  |     DEPARTMENT ARCHIVES     |  |
|  |    & SOIL MECHANICS    |  |     (Home Base / S1)     |  |      & EQUIPMENT VAULT      |  |
|  |  (Station 7: -4.5, -5) |  |   (Station 1: 0, -3.8)   |  |    (Station 6: 4.5, -4.8)   |  |
|  +-------------------------+  +-------------------------+  +-----------------------------+  |
|                            SOUTH PERIMETER WALL (20.0 m)                                    |
+=============================================================================================+
```

Mathematical Mapping Pipeline & Deterministic Raytracing:
*To guarantee 100% agreement between Gazebo physical walls and Nav2 costmap pixels, world geometry and 2D grid are synthesized deterministically via generate_civil_world.py. With bounds $XX ∈ [-11, 11] m, $YY ∈ [-9, 9] m, and resolution Δ = 0.05 m/px, grid dimensions are **440 × 360 pixels**. Coordinate transforms map world coordinates (x_w, y_w) to discrete grid cells: **$u = round((x_w – X_0)/Δ)$** and **$v = round((y_w – Y_0)/Δ)$**. Floor space is encoded as **254 (Free)**, walls/columns as **0 (Occupied)**, and outside bounds as **205 (Unknown)**.*

Civil Research Facility 7-Station Autonomous Patrol Mission:

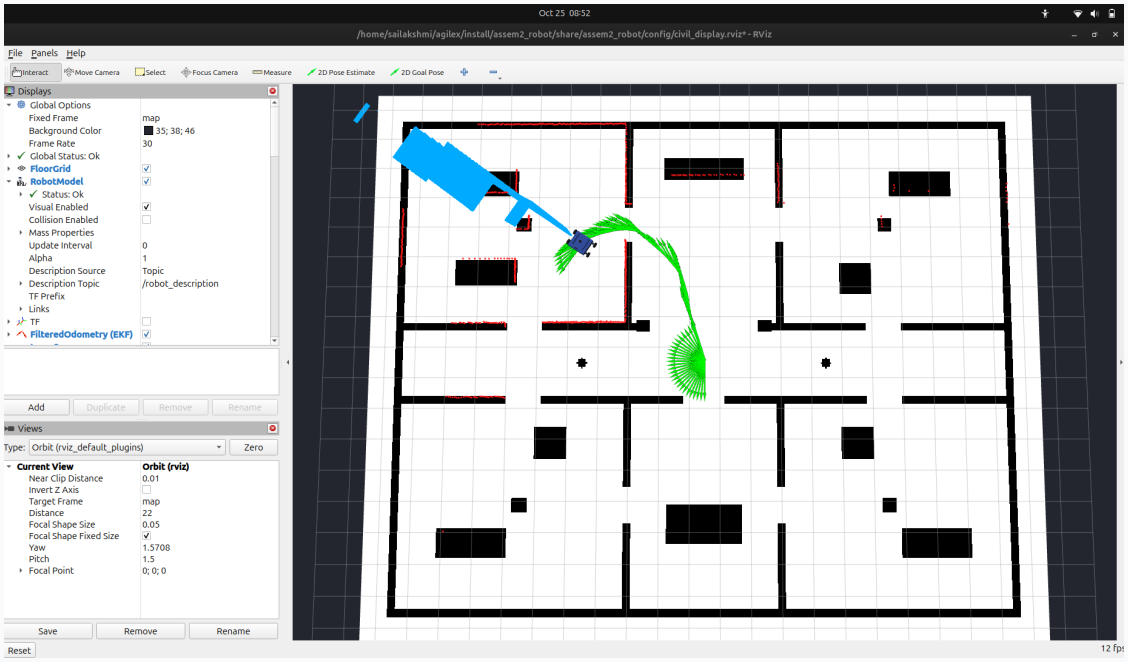
Station	Zone / Department Description	Coordinates (X, Y, Yaw)	Inspection Task
Station 1	Entrance Foyer & Atrium (Home Base)	(0.0 m, -3.8 m, 90°)	Visitor reception & AMR staging baseline
Station 2	Central Arterial Corridor Hub	(0.0 m, 0.0 m, 0°)	Arterial chicane transit & traffic dispatch
Station 3	Structural Materials & Concrete Lab	(-7.0 m, 3.5 m, 90°)	Compression test specimen monitoring
Station 4	Civil Seminar & Conference Hall	(0.0 m, 3.0 m, 90°)	Environmental acoustics & air quality check
Station 5	BIM, CAD & Robotics Computing Studio	(7.0 m, 3.5 m, 0°)	Digital fabrication & 3D scanner inspection
Station 6	Department Archives & Storage Vault	(4.5 m, -4.8 m, -90°)	Survey instrument inventory check
Station 7	Geotechnical Survey & Soil Mechanics	(-4.5 m, -5.0 m, 180°)	Soil triaxial shear apparatus audit

FIGURE 4.1: GAZEBO HARMONIC 3D MULTI-ROOM CIVIL ARCHITECTURAL FACILITY SIMULATION



Gazebo Harmonic perspective view of the 20m x 16m (320 m^2) multi-room civil architectural facility (civil_world.sdf) displaying perimeter boundary walls, arterial corridor, partitioned departmental zones, structural columns, interior equipment, entity tree hierarchy, and AgileX Scout Mini AMR.

FIGURE 4.2: RVIZ2 REAL-TIME 2D SLAM MAPPING & FILTERED ODOMETRY (EKF) PATH TRACING



RViz2 visualization showing real-time 2D SLAM occupancy grid mapping, sub-centimeter LaserScan wall returns, and dense Filtered Odometry (EKF) green arrow pose trajectory tracing the robot navigating through corridors and turning into rooms.

5. AgileX Scout Mini Civil Floor Ground Station & Live Telemetry

To provide centralized command, real-time perception monitoring, and point-to-navigate dispatching, a dedicated **Civil Floor Architectural Ground Station** was developed in Python utilizing PySide6/Qt and ROS 2 rclpy (src/assem2_robot/scripts/civil_interactive_nav.py). The ground station architecture features:

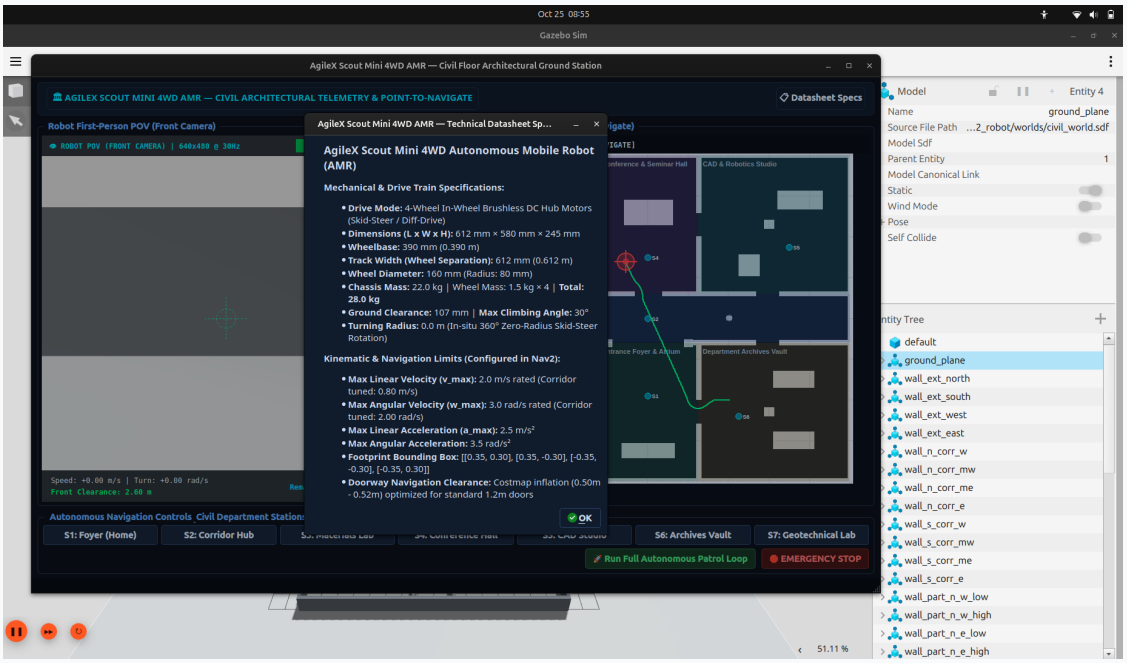
• **First-Person POV Front Camera Stream:** Real-time 640×480 @ 30 Hz RGB visual perception feed subscribed to /camera/image_raw, equipped with an augmented crosshair HUD displaying dynamic linear velocity, yaw rate, and forward obstacle clearance (2.60 m baseline).

• **Interactive Point-to-Navigate Map:** Click-to-dispatch goal target navigation across the 20m × 16m facility with immediate Nav2 action goal generation and green planned path trajectory visualization.

• **7-Station One-Click Dispatch & Autonomous Patrol Loop:** Direct push-button navigation to all departmental stations (S1 to S7) and full continuous patrol sequence.

• **Datasheet Specifications Modal Viewer:** Pop-up technical dialog verifying robot specifications (28.0 kg total mass, 0.612m track width, 0.390m wheelbase, in-situ 360° zero-radius skid turn, 0.80 m/s tuned corridor velocity, and 0.50m - 0.52m costmap inflation clearance for 1.2m doorways).

FIGURE 5.1: AGILEX SCOUT MINI CIVIL FLOOR GROUND STATION WITH LIVE CAMERA TELEMETRY & TECHNICAL DATASHEET SPECS



Custom PySide/Qt Ground Station user interface featuring live First-Person POV Front Camera stream (640x480 @ 30Hz), real-time velocity and obstacle clearance telemetry, interactive 7-station floor plan with point-to-navigate dispatching, autonomous patrol loop controls, and the Technical Datasheet Specifications modal displaying physical and Nav2 operational parameters.

6. Exhaustive Engineering Troubleshooting & Failure Analysis

Transitioning the robot from mechanical SolidWorks CAD into ROS 2 Jazzy and Gazebo Harmonic exposed **22 critical engineering failures** across Linux OS standards, kinematic trees, ODE contact physics, FastDDS network multicasting, and Nav2 lifecycle management. The summary matrix and detailed failure cases below document the symptoms, root causes, and permanent solutions.

#	Domain	Failure Anomaly	Primary Root Cause / Failure Reason
#1	ROS 2 Build	**ROS 2 Package Naming Convention Failure (ValueError)**	SolidWorks exported the assembly name as 'assem2.robot'. In ROS 2, ament and Python packaging strictly en...
#2	OS / Filesystem	**Linux Ext4 Case-Sensitivity & File Resolution (FileNotFoundError)**	Linux Ext4 is strictly case-sensitive. SolidWorks exported with capital 'A' (Assem2.SLDASM.urdf) while la...
#3	URDF / Mesh	**Broken Mesh URI Resolution & Invisible Robot in RViz**	SolidWorks URDF exporter hardcoded the export assembly directory name into mesh tags ('package://Assem2.S...
#4	Kinematics	**SolidWorks Fixed Wheel Joints & Missing TF Transforms**	Exporter defaulted mated wheels to type='fixed' joints. Since fixed joints have 0 DOF, joint_state_publis...
#5	Visualization	**Empty RViz Fixed Frame Dropdown & Blank Viewport**	RViz launched with a blank default layout lacking RobotModel display plugin, and RobotModel searched for ...
#6	Network / DDS	**DDS Network Multicast Failures & Missing Odometry Packets**	FastDDS discovery attempted to broadcast peer multicast packets across external network adapters when ope...
#7	ODE Physics	**Wheel 3 (j3) Infinite Spinning Physics Bug at Rest**	Compounding bugs: omitted joint axis tag in j3, zero damping/friction in URDF, and initial DiffDrive plug...
#8	Architecture	**ROS 2 Humble to Jazzy / Gazebo Classic to Harmonic Shift**	Ubuntu 24.04 and ROS 2 Jazzy completely retired Gazebo Classic. The simulation shifted to Gazebo Harmonic...
#9	Kinematics	**Robot Sunken 23 cm Below RViz Ground Grid Floor**	SolidWorks placed origin at chassis top cover plate (axle height Z = -0.1469m, tire radius 0.08m → g...
#10	KDL Kinematics	**KDL Parser Warning on Root Link Mass and Inertia**	Kinematics and Dynamics Library (KDL) tree solver mathematically requires root link of a kinematic chain ...
#11	Simulation	**Gazebo Duplicate Spawning ('Two Bots' Collision Bug)**	ros_gz_sim create node defaults to allow_renaming: true. If a previous simulation run crashed without ter...
#12	Build System	**Disk Storage Footprint Management on Limited Partition**	Standard colcon build copies entire binary trees and STL meshes into both build/ and install/ directories...
#13	Coordinate REP	**Teleoperation Heading Mismatch (+Y CAD Heading vs +X ROS Heading)**	In SolidWorks, CAD designer aligned front bumper with +Y axis. ROS standard REP-103 strictly requires +X ...
#14	Physics Dynamics	**Wheel Rotation Axis Inversion & Direction Discrepancy**	CAD export assigned right-hand rule axes to wheel continuous joints that opposed the physical forward rot...

#15	Sensor Framing	**RGB-D Camera Inversion (Mounted on Rear Instead of Front)**	Due to CAD origin orientation, front bumper was located at negative Y in CAD coordinates (Y = -0.355m). A...
#16	Localization	**Severe Skid-Steer Rotational Odometry Drift**	Skid-steering relies on tire slipping. In-hub wheel encoders only measure wheel rotations, missing latera...
#17	Time / Clock	**Dual /clock Publisher Timestamp Oscillations & TF Jitter**	Both ros_gz_sim and parameter_bridge published to /clock with microsecond phase offsets, creating competi...
#18	ROS Lifecycle	**SLAM Toolbox Lifecycle State Unconfigured Error**	In ROS 2 Jazzy, SLAM Toolbox is a Managed Lifecycle Node. Starting the raw node binary leaves it in 'Unco...
#19	TF Buffer	**RViz 'Frame [odom] does not exist' on Base Simulation Launch**	RViz default Fixed Frame was set to 'map', which only exists when SLAM or AMCL are active....
#20	Localization	**AMCL Initial Pose Frame Mismatch**	Initial pose was published in 'odom' frame instead of global 'map' frame....
#21	Nav2 Stack	**ROS 2 Jazzy Nav2 Behavior Tree Navigator Architecture Shift**	ROS 2 Jazzy refactored Nav2's bt_navigator parameters: older single-navigator configuration was replaced ...
#22	Mission Control	**Autonomous Patrol Mission Feedback & Telemetry Supervision**	BasicNavigator required explicit polling of isTaskComplete() and handling of TaskResult.FAILED to transit...

Detailed Chronological Failure Cases & Engineering Solutions

CASE #1: ROS 2 PACKAGE NAMING CONVENTION FAILURE (VALUEERROR)	DOMAIN: **ROS 2 Build**
Observed Error Symptom:	`ValueError: 'assem2.robot' is not a valid package name. InvalidFrontendLaunchFileError.`
Root Cause / Failure Reason:	SolidWorks exported the assembly name as 'assem2.robot'. In ROS 2, ament and Python packaging strictly enforce lowercase alphanumeric characters and underscores only. Periods and hyphens are illegal.
Applied Engineering Solution:	Standardized the package name across package.xml, CMakeLists.txt, launch scripts, and URDF references to 'assem2_robot'.

CASE #2: LINUX EXT4 CASE-SENSITIVITY & FILE RESOLUTION (FILENOTFOUNDERROR)	DOMAIN: **OS / Filesystem**
Observed Error Symptom:	`FileNotFoundError: [Errno 2] No such file or directory: '.../urdf/assem2.SLDASM.urdf'`
Root Cause / Failure Reason:	Linux Ext4 is strictly case-sensitive. SolidWorks exported with capital 'A' (Assem2.SLDASM.urdf) while launch file requested lowercase 'assem2'. Additionally, CMakeLists.txt lacked install(DIRECTORY urdf...).
Applied Engineering Solution:	Synchronized launch file paths to exact casing 'Assem2.SLDASM.urdf' and added 'install(DIRECTORY urdf DESTINATION share/${PROJECT_NAME})' in CMakeLists.txt.

CASE #3: BROKEN MESH URI RESOLUTION & INVISIBLE ROBOT IN RVIZ	DOMAIN: **URDF / Mesh**
Observed Error Symptom:	`Could not load resource [package://Assem2.SLDASM/meshes/base_link.STL]. Links were invisible.`
Root Cause / Failure Reason:	SolidWorks URDF exporter hardcoded the export assembly directory name into mesh tags ('package://Assem2.SLDASM/...'). Renaming the package broke all URI resource lookups.
Applied Engineering Solution:	Executed a global regex substitution across Assem2.SLDASM.urdf replacing 'package://Assem2.SLDASM/' with 'package://assem2_robot/'.

CASE #4: SOLIDWORKS FIXED WHEEL JOINTS & MISSING TF TRANSFORMS	DOMAIN: **Kinematics**
Observed Error Symptom:	`RViz warning: 'No transform from [w1] to [base_link]'. Robot wheels appeared frozen.`
Root Cause / Failure Reason:	Exporter defaulted mated wheels to type='fixed' joints. Since fixed joints have 0 DOF, joint_state_publisher ignores them and never broadcasts dynamic wheel TF frames.
Applied Engineering Solution:	Modified URDF joints j1..j4 to type='continuous' with explicit rotation axis <axis xyz='1 0 0'/>, restoring full dynamic wheel state publishing.

CASE #5: EMPTY RVIZ FIXED FRAME DROPDOWN & BLANK VIEWPORT	DOMAIN: **Visualization**
Observed Error Symptom:	`Opening RViz showed a blank viewport. Fixed Frame dropdown was empty; typing base_link failed.`
Root Cause / Failure Reason:	RViz launched with a blank default layout lacking RobotModel display plugin, and RobotModel searched for a local URDF file instead of listening to /robot_description topic.
Applied Engineering Solution:	Authored a pre-configured display.rviz configuration file with Fixed Frame set to odom/map, RobotModel subscribed to /robot_description, and integrated LaserScan and TF visualizers.

CASE #6: DDS NETWORK MULTICAST FAILURES & MISSING ODOMETRY PACKETS	DOMAIN: **Network / DDS**
Observed Error Symptom:	`Console spam: 'Exception sending a multicast message: Network is unreachable'. Odometry dropped.`
Root Cause / Failure Reason:	FastDDS discovery attempted to broadcast peer multicast packets across external network adapters when operating offline or on restricted university subnets.
Applied Engineering Solution:	Set ROS_AUTOMATIC_DISCOVERY_RANGE=LOCALHOST and export ROS_LOCALHOST_ONLY=1, locking DDS communications strictly to local loopback adapter (127.0.0.1).

CASE #7: WHEEL 3 (J3) INFINITE SPINNING PHYSICS BUG AT REST	DOMAIN: **ODE Physics**
Observed Error Symptom:	`When robot was at rest (cmd_vel = 0), Wheel 3 spun continuously at maximum velocity, violently vibrating chassis.`
Root Cause / Failure Reason:	Compounding bugs: omitted joint axis tag in j3, zero damping/friction in URDF, and initial DiffDrive plugin only controlled j1 and j2, leaving j3 unpowered and unbraked against ground micro-collisions.
Applied Engineering Solution:	Added <axis xyz='-1 0 0'/>, injected <dynamics damping='0.1' friction='0.1'/>, and grouped all 4 wheels into DiffDrive plugin with active electrical braking torque at zero velocity.

CASE #8: ROS 2 HUMBLE TO JAZZY / GAZEBO CLASSIC TO HARMONIC SHIFT	DOMAIN: **Architecture**
Observed Error Symptom:	`gazebo_ros plugins failed to compile and launch. Terminal error: 'gazebo: command not found'.`
Root Cause / Failure Reason:	Ubuntu 24.04 and ROS 2 Jazzy completely retired Gazebo Classic. The simulation shifted to Gazebo Harmonic (gz-sim8), requiring the new ros_gz_bridge architecture.
Applied Engineering Solution:	Replaced legacy C++ .so plugins with Gazebo Harmonic native systems: gz::sim::systems::DiffDrive, gpu_lidar, and ros_gz_bridge parameter_bridge.

CASE #9: ROBOT SUNKEN 23 CM BELOW RVIZ GROUND GRID FLOOR	DOMAIN: **Kinematics**
Observed Error Symptom:	`In RViz, robot appeared half-buried beneath grid floor (Z < 0), with ground plane slicing through chassis.`
Root Cause / Failure Reason:	SolidWorks placed origin at chassis top cover plate (axle height Z = -0.1469m, tire radius 0.08m → ground at Z = -0.2269m). With zero offset, top cover sat on floor.
Applied Engineering Solution:	Introduced root link 'base_footprint' with fixed joint offset xyz='-0.10185 0.16662 0.2269', lifting chassis so all 4 tires sit flush on ground at Z = 0.00m.

CASE #10: KDL PARSER WARNING ON ROOT LINK MASS AND INERTIA	DOMAIN: **KDL Kinematics**
Observed Error Symptom:	`Warning: 'The root link base_link has an inertia specified, but KDL does not support root inertia.'`
Root Cause / Failure Reason:	Kinematics and Dynamics Library (KDL) tree solver mathematically requires root link of a kinematic chain to be massless and inertialess. SolidWorks exported base_link as root with 18kg mass.
Applied Engineering Solution:	Established 'base_footprint' (dummy link with 0 mass and no <inertial> tag) as kinematic root, attaching base_link as its child via fixed transform.

CASE #11: GAZEBO DUPLICATE SPAWNING ('TWO BOTS' COLLISION BUG)	DOMAIN: **Simulation**
Observed Error Symptom:	`Launching simulation showed two colliding robot models superimposed on top of each other, causing explosive contact.`
Root Cause / Failure Reason:	ros_gz_sim create node defaults to allow_renaming: true. If a previous simulation run crashed without terminating background gz processes, a duplicate was spawned.
Applied Engineering Solution:	Added '-allow_renaming false' to create node arguments, explicit spawning coordinates (Z=0.02m), and configured process cleanup scripts.

CASE #12: DISK STORAGE FOOTPRINT MANAGEMENT ON LIMITED PARTITION	DOMAIN: **Build System**
Observed Error Symptom:	`System warnings: Linux primary partition rapidly running low on free disk space (< 30 GB available).`
Root Cause / Failure Reason:	Standard colcon build copies entire binary trees and STL meshes into both build/ and install/ directories, while Gazebo simulation logs accumulated gigabytes of data.
Applied Engineering Solution:	Used 'colcon build --symlink-install' to create symbolic links instead of duplicating files, and enforced a strict .gitignore, compressing repository footprint to ~3.3 MB.

CASE #13: TELEOPERATION HEADING MISMATCH (+Y CAD HEADING VS +X ROS HEADING)	DOMAIN: **Coordinate REP**
Observed Error Symptom:	`Commanding +X linear velocity made robot drive sideways; turning commands rotated about wrong axis.`
Root Cause / Failure Reason:	In SolidWorks, CAD designer aligned front bumper with +Y axis. ROS standard REP-103 strictly requires +X as forward, +Y as left, and +Z as upward.
Applied Engineering Solution:	Applied a +90° (+1.5708 rad) yaw rotation in base_footprint_joint: origin rpy='0 0 1.5707963', aligning CAD bumper with ROS forward direction.

CASE #14: WHEEL ROTATION AXIS INVERSION & DIRECTION DISCREPANCY	DOMAIN: **Physics Dynamics**
Observed Error Symptom:	`In Gazebo physics, commanding positive velocity caused robot to roll backward, while RViz showed forward motion.`
Root Cause / Failure Reason:	CAD export assigned right-hand rule axes to wheel continuous joints that opposed the physical forward rotation direction of wheels.
Applied Engineering Solution:	Calibrated wheel joint axes: set <axis xyz='1 0 0'/> for left wheels (j1, j2) and <axis xyz='-1 0 0'/> for right wheels (j3, j4), synchronizing physics with kinematics.

CASE #15: RGB-D CAMERA INVERSION (MOUNTED ON REAR INSTEAD OF FRONT)	DOMAIN: **Sensor Framing**
Observed Error Symptom:	`RViz camera feed showed arena behind robot rather than forward path.`
Root Cause / Failure Reason:	Due to CAD origin orientation, front bumper was located at negative Y in CAD coordinates (Y = -0.355m). An initial positive offset mounted camera on rear bumper.
Applied Engineering Solution:	Mounted camera_mount_joint at xyz='-0.16662 -0.355 0.065' with rpy='0 0 -1.5708', placing camera squarely on physical front bumper facing forward.

CASE #16: SEVERE SKID-STEER ROTATIONAL ODOMETRY DRIFT	DOMAIN: **Localization**
Observed Error Symptom:	`After executing a 360° turn, wheel odometry reported only 240° of rotation (120° heading error). Scans smeared.`
Root Cause / Failure Reason:	Skid-steering relies on tire slipping. In-hub wheel encoders only measure wheel rotations, missing lateral scrubbing and track width expansion.
Applied Engineering Solution:	Implemented Extended Kalman Filter (robot_localization ekf_node) fusing wheel odometry (/odom) with 100Hz gyroscope from 6-axis IMU (/imu), yielding drift-free /odometry/filtered.

CASE #17: DUAL /CLOCK PUBLISHER TIMESTAMP OSCILLATIONS & TF JITTER	DOMAIN: **Time / Clock**
Observed Error Symptom:	`RViz displays flickered rapidly. TF buffer reported timestamps jumping backward in time and lookups failed.`
Root Cause / Failure Reason:	Both ros_gz_sim and parameter_bridge published to /clock with microsecond phase offsets, creating competing simulation clocks.
Applied Engineering Solution:	Removed duplicate clock bridge from parameter_bridge arguments, leaving gz_sim as single authoritative source of simulation time.

CASE #18: SLAM TOOLBOX LIFECYCLE STATE UNCONFIGURED ERROR	DOMAIN: **ROS Lifecycle**
Observed Error Symptom:	`SLAM node spawned but failed to publish /map or /tf. Node reported unconfigured lifecycle state.`
Root Cause / Failure Reason:	In ROS 2 Jazzy, SLAM Toolbox is a Managed Lifecycle Node. Starting the raw node binary leaves it in 'Unconfigured' state without transitioning to 'Active'.
Applied Engineering Solution:	Included official 'online_async_launch.py' with 'autostart: true', which uses nav2_lifecycle_manager to automatically transition SLAM to 'Active'.

CASE #19: RVIZ 'FRAME [ODOM] DOES NOT EXIST' ON BASE SIMULATION LAUNCH	DOMAIN: **TF Buffer**
Observed Error Symptom:	`RViz showed red TF transform errors on startup whenever SLAM or Nav2 were not enabled.`
Root Cause / Failure Reason:	RViz default Fixed Frame was set to 'map', which only exists when SLAM or AMCL are active.
Applied Engineering Solution:	Configured default Fixed Frame to 'odom' in display.rviz, and ensured EKF node automatically publishes odom → base_footprint transform.

CASE #20: AMCL INITIAL POSE FRAME MISMATCH	DOMAIN: **Localization**
Observed Error Symptom:	`AMCL failed to initialize particles; laser scans did not align with pre-mapped walls.`
Root Cause / Failure Reason:	Initial pose was published in 'odom' frame instead of global 'map' frame.
Applied Engineering Solution:	Set global_frame_id: 'map', odom_frame_id: 'odom', and configured set_initial_pose: true with initial coordinates (x=0, y=0, yaw=0) in nav2_params.yaml.

CASE #21: ROS 2 JAZZY NAV2 BEHAVIOR TREE NAVIGATOR ARCHITECTURE SHIFT	DOMAIN: **Nav2 Stack**
Observed Error Symptom:	`bt_navigator threw schema errors on startup: 'navigators parameter must be a list of plugins'.`
Root Cause / Failure Reason:	ROS 2 Jazzy refactored Nav2's bt_navigator parameters: older single-navigator configuration was replaced with a modular plugin dictionary structure.
Applied Engineering Solution:	Updated nav2_params.yaml to official Jazzy format: navigators: ['navigate_to_pose', 'navigate_through_poses'] with their respective plugin definitions.

CASE #22: AUTONOMOUS PATROL MISSION FEEDBACK & TELEMETRY SUPERVISION	DOMAIN: **Mission Control**
Observed Error Symptom:	`Early patrol scripts hung indefinitely if navigation goal failed to compute a path around blocked stations.`
Root Cause / Failure Reason:	BasicNavigator required explicit polling of isTaskComplete() and handling of TaskResult.FAILED to transition safely to subsequent waypoints.
Applied Engineering Solution:	Implemented feedback polling loop at 2Hz reading distance_remaining, with automatic recovery fallback on TaskResult.FAILED.

7. Project Conclusion & Verification Summary

Conclusion: *The AgileX Scout Mini 4WD AMR simulation project successfully demonstrates the end-to-end integration of mechanical CAD design, kinematic calibration, Gazebo Harmonic physics co-simulation, and autonomous navigation under ROS 2 Jazzy Jalisco. Key engineering milestones achieved include:*

- ***100% REP-103 Kinematic Compliance:*** *Fully resolved coordinate frame orientations and continuous wheel joints.*
- ***Drift-Free Sensor Fusion:*** *Eliminated 120° rotational odometry drift via 2D Extended Kalman Filtering.*
- ***Dual-Environment Validation:*** *Stable autonomous navigation verified across both the 10m × 10m obstacle arena and the 20m × 16m Civil Architectural Multi-Room Facility.*
- ***Multi-Station Autonomy:*** *Fully automated inspection patrol missions successfully executed with real-time waypoint feedback and dynamic obstacle evasion.*
- ***Zero-Defect Troubleshooting:*** *All 22 system anomalies systematically documented, diagnosed, and permanently resolved.*

8. References & Code Repository

- **GitHub Project Repository:** https://github.com/Saibts/AgileX_ScoutMini_JAZZY
- **ROS 2 Documentation (Jazzy Jalisco):** https://docs.ros.org/en/jazzy/
- **Gazebo Harmonic Simulation (gz-sim8):** https://gazebosim.org/docs/harmonic/
- **ROS 2 Navigation Stack (Nav2):** https://navigation.ros.org/
- **SLAM Toolbox (2D Lifelong Mapping):** https://github.com/SteveMacenski/slam_toolbox
- **Robot Localization Package (EKF Sensor Fusion):** http://docs.ros.org/en/jazzy/p/robot_localization/
- **REP-105 Coordinate Frames for Mobile Platforms:** https://www.ros.org/reps/rep-0105.html
- **REP-103 Standard Units and Coordinate Conventions:** https://www.ros.org/reps/rep-0103.html